

In-Class Exercises: Generalized Convexity Classes

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Definitions Reference

All functions below are defined on a nonempty open convex set $S \subseteq \mathbb{R}^n$ and are differentiable unless otherwise stated. Let $\mathbf{x}_1, \mathbf{x}_2 \in S$ and $\lambda \in (0, 1)$.

1. Convex

$$f(\lambda \mathbf{x}_1 + (1 - \lambda) \mathbf{x}_2) \leq \lambda f(\mathbf{x}_1) + (1 - \lambda) f(\mathbf{x}_2)$$

Equivalent (differentiable): $f(\mathbf{x}_2) \geq f(\mathbf{x}_1) + \nabla f(\mathbf{x}_1)^T (\mathbf{x}_2 - \mathbf{x}_1)$ for all $\mathbf{x}_1, \mathbf{x}_2 \in S$.

Equivalent (twice differentiable): $\nabla^2 f(\mathbf{x}) \succeq 0$ for all $\mathbf{x} \in S$.

2. Strictly Convex

$$\text{For } \mathbf{x}_1 \neq \mathbf{x}_2: \quad f(\lambda \mathbf{x}_1 + (1 - \lambda) \mathbf{x}_2) < \lambda f(\mathbf{x}_1) + (1 - \lambda) f(\mathbf{x}_2)$$

Sufficient (not necessary): $\nabla^2 f(\mathbf{x}) \succ 0$ for all $\mathbf{x} \in S$.

3. μ -Strongly Convex ($\mu > 0$)

$$g(\mathbf{x}) := f(\mathbf{x}) - \mu \|\mathbf{x}\|^2 \text{ is convex.}$$

Equivalent (twice differentiable): $\nabla^2 f(\mathbf{x}) \succeq 2\mu I_n$ for all $\mathbf{x} \in S$.

Consequence: Strongly convex $\Rightarrow$ Strictly convex $\Rightarrow$ Convex.

4. Pseudoconvex

$$\nabla f(\mathbf{x}_1)^T (\mathbf{x}_2 - \mathbf{x}_1) \geq 0 \quad \Rightarrow \quad f(\mathbf{x}_2) \geq f(\mathbf{x}_1)$$

Intuition: “The gradient never lies.” If the gradient says the function doesn’t decrease in a direction, it really doesn’t.

Key property: $\nabla f(\bar{\mathbf{x}}) = 0 \Rightarrow \bar{\mathbf{x}}$ is a global minimizer.

5. Strictly Quasiconvex

$$\text{For } \mathbf{x}_1 \neq \mathbf{x}_2 \text{ with } f(\mathbf{x}_1) \neq f(\mathbf{x}_2): \quad f(\lambda \mathbf{x}_1 + (1 - \lambda) \mathbf{x}_2) < \max\{f(\mathbf{x}_1), f(\mathbf{x}_2)\}$$

Intuition: If one endpoint is strictly higher, the interior of the segment stays strictly below it.

6. Quasiconvex

$$f(\lambda \mathbf{x}_1 + (1 - \lambda) \mathbf{x}_2) \leq \max\{f(\mathbf{x}_1), f(\mathbf{x}_2)\}$$

Equivalent: All lower level sets $\{\mathbf{x} \in S : f(\mathbf{x}) \leq \alpha\}$ are convex.

First-order: $f(\mathbf{x}_2) \leq f(\mathbf{x}_1) \Rightarrow \nabla f(\mathbf{x}_1)^T (\mathbf{x}_2 - \mathbf{x}_1) \leq 0$.

The Generalized Convexity Hierarchy (for differentiable functions on open convex sets):

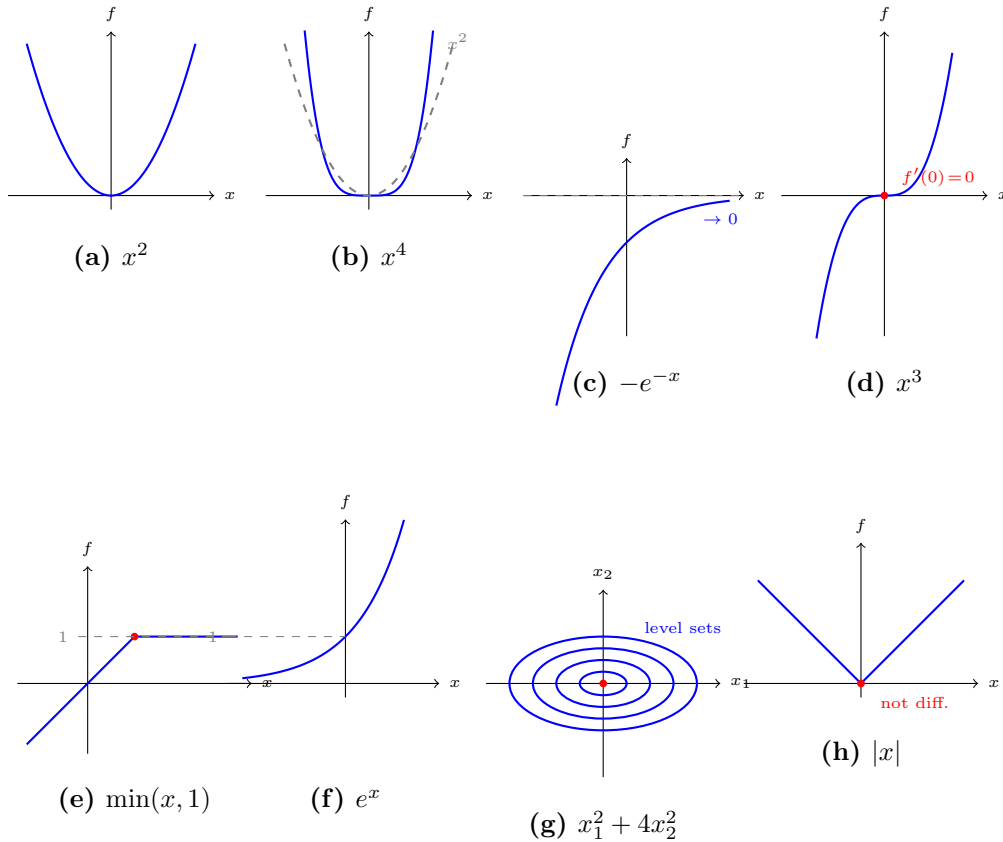
$$\text{Str. Cvx} \subsetneq \text{Strictly Cvx} \subsetneq \text{Cvx} \subsetneq \text{Pseudocvx} \subsetneq \text{Str. QC} \subsetneq \text{QC}$$

Each inclusion is **strict**: there exist functions in each class that do not belong to the smaller class.

1 Part I: Classification (20 min)

For each function below, determine which convexity classes it belongs to. Check **all** that apply. Assume $S = \mathbb{R}$ (or $\mathbb{R}^n$) unless otherwise stated.

Function Gallery



(a) $f(x) = x^2$ on $\mathbb{R}$.

Strongly Cvx	Strictly Cvx	Convex	Pseudocvx	Strictly QC	Quasicvx
☐	☐	☐	☐	☐	☐

(b) $f(x) = x^4$ on $\mathbb{R}$.

Strongly Cvx	Strictly Cvx	Convex	Pseudocvx	Strictly QC	Quasicvx
☐	☐	☐	☐	☐	☐

(c) $f(x) = -e^{-x}$ on $\mathbb{R}$.

Strongly Cvx	Strictly Cvx	Convex	Pseudocvx	Strictly QC	Quasicvx
☐	☐	☐	☐	☐	☐

(d) $f(x) = x^3$ on $\mathbb{R}$.

Strongly Cvx	Strictly Cvx	Convex	Pseudocvx	Strictly QC	Quasicvx
☐	☐	☐	☐	☐	☐

(e) $f(x) = \min(x, 1)$ on $\mathbb{R}$.

Strongly Cvx	Strictly Cvx	Convex	Pseudocvx	Strictly QC	Quasicvx
☐	☐	☐	☐	☐	☐

(f) $f(x) = e^x$ on $\mathbb{R}$.

Strongly Cvx	Strictly Cvx	Convex	Pseudocvx	Strictly QC	Quasicvx
☐	☐	☐	☐	☐	☐

(g) $f(x_1, x_2) = x_1^2 + 4x_2^2$ on $\mathbb{R}^2$.

Strongly Cvx	Strictly Cvx	Convex	Pseudocvx	Strictly QC	Quasicvx
☐	☐	☐	☐	☐	☐

(h) $f(x) = |x|$ on $\mathbb{R}$. (Note: not differentiable at $x = 0$.)

Strongly Cvx	Strictly Cvx	Convex	Pseudocvx	Strictly QC	Quasicvx
☐	☐	☐	☐	☐	☐

2 Part II: Proof Exercises (25 min)

- (a) **Prove** that every convex function is pseudoconvex.

Hint: Start from the first-order characterization of convexity and show it implies the pseudoconvexity condition.

- (b) **Prove** that every pseudoconvex function is strictly quasiconvex.

Hint: Use the contrapositive of pseudoconvexity. Take $\mathbf{x}_1, \mathbf{x}_2$ with $f(\mathbf{x}_1) \neq f(\mathbf{x}_2)$ and show that $f(\lambda \mathbf{x}_1 + (1 - \lambda) \mathbf{x}_2) < \max\{f(\mathbf{x}_1), f(\mathbf{x}_2)\}$.

- (c) **Prove** that if f is pseudoconvex and $\nabla f(\bar{\mathbf{x}}) = 0$, then $\bar{\mathbf{x}}$ is a global minimizer.

Hint: This should be a short proof (2–3 lines).

- (d) **Prove or disprove:** If f is quasiconvex and $\nabla f(\bar{\mathbf{x}}) = 0$, then $\bar{\mathbf{x}}$ is a local minimizer.

3 Part III: Counterexamples — Strict Separation (20 min)

For each pair below, give an explicit example of a function that belongs to the **larger** class but *not* to the **smaller** class. Justify your answer.

- (a) Find a function that is **strictly convex** but *not* **strongly convex**.

Hint: Think about a function whose second derivative goes to 0.

- (b) Find a function that is **convex** but *not* **strictly convex**.

- (c) Find a function that is **pseudoconvex** but *not* **convex**.

Hint: The function $f(x) = -e^{-x}$ is concave. Can a concave function be pseudoconvex?

- (d) Find a function that is **strictly quasiconvex** but *not* **pseudoconvex**.

Hint: Consider $f(x) = x^3$. Check what happens at $x = 0$.

- (e) Find a function that is **quasiconvex** but *not* **strictly quasiconvex**.

Hint: Think about a function that is constant on an interval.

4 Part IV: True/False and Relationships (10 min)

Mark each statement as True or False. If false, give a counterexample.

- (a) Every strongly convex function has a unique global minimizer on $\mathbb{R}^n$. ☐ True ☐ False
- (b) If f is quasiconvex and has a stationary point, that point is a global minimizer. ☐ True ☐ False
- (c) If f is pseudoconvex, then $-f$ is pseudoconvex. ☐ True ☐ False
- (d) The sum of two quasiconvex functions is quasiconvex. ☐ True ☐ False
- (e) Every linear function $f(\mathbf{x}) = \mathbf{c}^T \mathbf{x}$ is simultaneously convex, pseudoconvex, and quasiconvex. ☐ True ☐ False
- (f) If f is strictly quasiconvex and $\bar{\mathbf{x}}$ is a local minimizer, then $\bar{\mathbf{x}}$ is the unique global minimizer. ☐ True ☐ False

Solutions

Part I Solutions: Classification

- (a) $f(x) = x^2$: $f''(x) = 2 > 0$ everywhere, so f is 1-strongly convex ($\nabla^2 f \geq 2\mu I$ with $\mu = 1$). By the hierarchy, it is also strictly convex, convex, pseudoconvex, strictly quasiconvex, and quasiconvex. **All six boxes checked.**
- (b) $f(x) = x^4$: $f''(x) = 12x^2$. At $x = 0$, $f''(0) = 0$, so f is *not* strongly convex ($\nabla^2 f \not\geq 2\mu I$ for any $\mu > 0$). However, $f''(x) \geq 0$ everywhere, so f is convex. It is strictly convex because $f(\lambda x_1 + (1 - \lambda)x_2) < \lambda f(x_1) + (1 - \lambda)f(x_2)$ for $x_1 \neq x_2$ (the function is not affine on any interval). By hierarchy: **all except strongly convex.**
- (c) $f(x) = -e^{-x}$: $f'(x) = e^{-x} > 0$ (strictly increasing), $f''(x) = -e^{-x} < 0$ (strictly concave). Not convex, not strictly convex, not strongly convex. For pseudoconvexity: $f'(x_1)(x_2 - x_1) \geq 0$ means $x_2 \geq x_1$ (since $f'(x_1) > 0$), which gives $f(x_2) \geq f(x_1)$ (strictly increasing). So f is pseudoconvex, hence also strictly quasiconvex and quasiconvex. **Pseudoconvex, Str. QC, QC only.**
- (d) $f(x) = x^3$: $f'(x) = 3x^2$, $f''(x) = 6x$. At $x = 0$: $f'(0) = 0$ but $x = 0$ is *not* a global minimizer (function goes to $-\infty$). So f is *not pseudoconvex* (critical point that is not a global min). However, f is strictly increasing on $(-\infty, 0)$ and $(0, \infty)$ with only a single inflection at 0. It is strictly quasiconvex: if $f(x_1) \neq f(x_2)$, then the interior of $[x_1, x_2]$ stays strictly below the max. It is also quasiconvex (level sets $\{x : x^3 \leq \alpha\} = (-\infty, \alpha^{1/3}]$ are convex). Not convex ($f'' < 0$ for $x < 0$). **Str. QC and QC only.**
- (e) $f(x) = \min(x, 1)$: This is concave (not convex). Not differentiable at $x = 1$, but continuous. Level sets $\{x : f(x) \leq \alpha\}$: for $\alpha < 1$, this is $(-\infty, \alpha]$ (convex); for $\alpha \geq 1$, this is $\mathbb{R}$ (convex). So f is quasiconvex. However, f is constant ($= 1$) on $[1, 2]$. If $x_1 = 1, x_2 = 2$: $f(x_1) = f(x_2) = 1$ so the strict QC condition doesn't apply (requires $f(x_1) \neq f(x_2)$). But consider $x_1 = 0, x_2 = 2$: $f(0) = 0 \neq f(2) = 1$, and $f(0.5 \cdot 0 + 0.5 \cdot 2) = f(1) = 1 = \max\{0, 1\}$, *not* strictly less. So f is *not* strictly quasiconvex. **QC only.**
- (f) $f(x) = e^x$: $f''(x) = e^x > 0$ everywhere. But e^x is not bounded below by a quadratic: $e^x - \mu x^2$ is convex only if $e^x \geq 2\mu$ for all x , which fails as $x \rightarrow -\infty$. So f is *not* strongly convex. It is strictly convex ($f'' > 0$), hence convex, pseudoconvex, strictly QC, and QC. **All except strongly convex.**
- (g) $f(x_1, x_2) = x_1^2 + 4x_2^2$: $\nabla^2 f = \text{diag}(2, 8) \succeq 2 \cdot 1 \cdot I_2$. So f is 1-strongly convex. **All six boxes checked.**
- (h) $f(x) = |x|$: Convex (triangle inequality). Strictly convex (not affine on any interval). Not strongly convex ($f''(x) = 0$ for $x \neq 0$). Not differentiable at $x = 0$, so pseudoconvexity (which requires differentiability) does not apply in the standard sense. Level sets are convex, so QC and strictly QC (by strict convexity). **Strictly Cvx, Convex, Str. QC, QC.** Pseudoconvex: N/A (not differentiable).

Part II Solutions: Proofs

(a) Proof: Convex $\Rightarrow$ Pseudoconvex.

Let f be convex and differentiable on S . The first-order characterization of convexity gives:

$$f(\mathbf{x}_2) \geq f(\mathbf{x}_1) + \nabla f(\mathbf{x}_1)^T(\mathbf{x}_2 - \mathbf{x}_1) \quad \text{for all } \mathbf{x}_1, \mathbf{x}_2 \in S.$$

Now suppose $\nabla f(\mathbf{x}_1)^T(\mathbf{x}_2 - \mathbf{x}_1) \geq 0$. Then:

$$f(\mathbf{x}_2) \geq f(\mathbf{x}_1) + \underbrace{\nabla f(\mathbf{x}_1)^T(\mathbf{x}_2 - \mathbf{x}_1)}_{\geq 0} \geq f(\mathbf{x}_1).$$

This is exactly the definition of pseudoconvexity. □

(b) Proof: Pseudoconvex $\Rightarrow$ Strictly Quasiconvex.

Let f be pseudoconvex on S . Take $\mathbf{x}_1, \mathbf{x}_2 \in S$ with $f(\mathbf{x}_1) \neq f(\mathbf{x}_2)$. WLOG, assume $f(\mathbf{x}_2) < f(\mathbf{x}_1)$.

By the contrapositive of pseudoconvexity: $f(\mathbf{x}_2) < f(\mathbf{x}_1) \Rightarrow \nabla f(\mathbf{x}_1)^T(\mathbf{x}_2 - \mathbf{x}_1) < 0$.

For $\lambda \in (0, 1)$, let $\mathbf{x}_\lambda = \lambda \mathbf{x}_1 + (1 - \lambda) \mathbf{x}_2$. Then $\mathbf{x}_\lambda - \mathbf{x}_1 = (1 - \lambda)(\mathbf{x}_2 - \mathbf{x}_1)$, so:

$$\nabla f(\mathbf{x}_1)^T(\mathbf{x}_\lambda - \mathbf{x}_1) = (1 - \lambda) \underbrace{\nabla f(\mathbf{x}_1)^T(\mathbf{x}_2 - \mathbf{x}_1)}_{< 0} < 0.$$

By the contrapositive of pseudoconvexity again (applied with $\mathbf{x}_\lambda$ in place of $\mathbf{x}_2$):

$$\nabla f(\mathbf{x}_1)^T(\mathbf{x}_\lambda - \mathbf{x}_1) < 0 \quad \Rightarrow \quad f(\mathbf{x}_\lambda) < f(\mathbf{x}_1) = \max\{f(\mathbf{x}_1), f(\mathbf{x}_2)\}.$$

This is exactly the strict quasiconvexity condition. □

(c) Proof: Pseudoconvex + $\nabla f(\bar{\mathbf{x}}) = 0 \Rightarrow \bar{\mathbf{x}}$ is global min.

For any $\mathbf{x} \in S$:

$$\nabla f(\bar{\mathbf{x}})^T(\mathbf{x} - \bar{\mathbf{x}}) = \mathbf{0}^T(\mathbf{x} - \bar{\mathbf{x}}) = 0 \geq 0.$$

By pseudoconvexity: $\nabla f(\bar{\mathbf{x}})^T(\mathbf{x} - \bar{\mathbf{x}}) \geq 0 \Rightarrow f(\mathbf{x}) \geq f(\bar{\mathbf{x}})$.

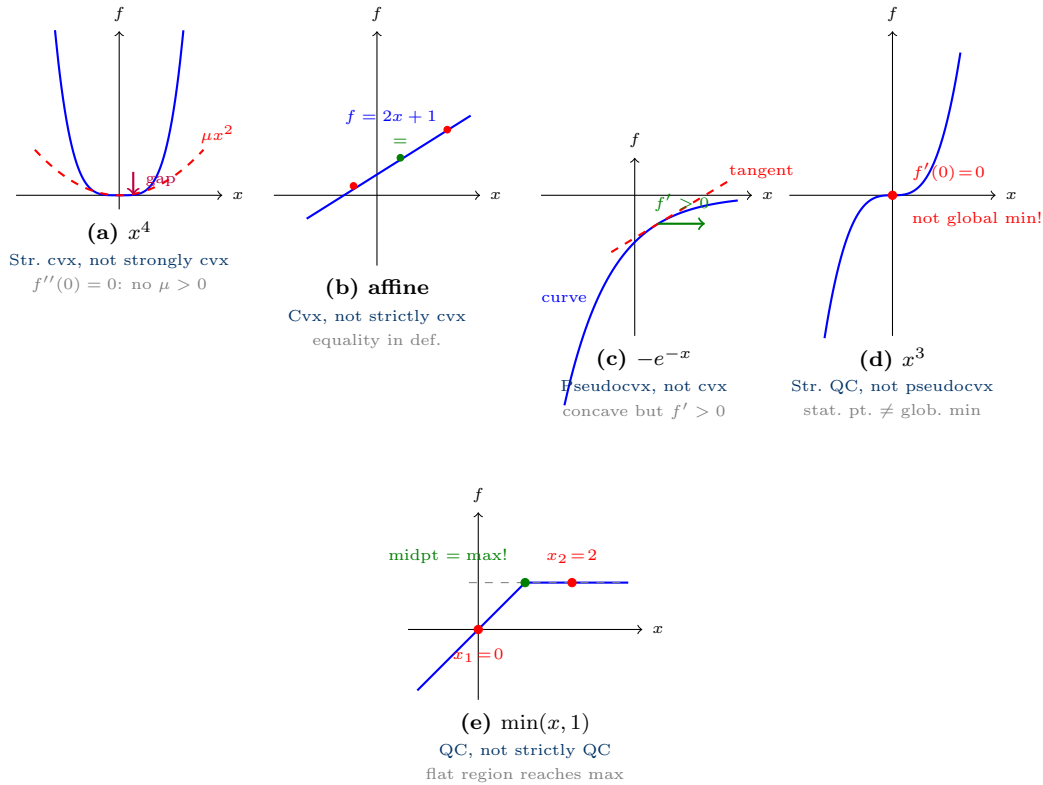
Since $\mathbf{x}$ was arbitrary, $\bar{\mathbf{x}}$ is a global minimizer. □

(d) Disprove: $f(x) = x^3$ is quasiconvex (level sets $(-\infty, \alpha^{1/3}]$ are convex). At $\bar{\mathbf{x}} = 0$: $f'(0) = 3(0)^2 = 0$, so $\bar{\mathbf{x}} = 0$ is a stationary point. But $f(-1) = -1 < 0 = f(0)$, so $\bar{\mathbf{x}} = 0$ is **not** a local minimizer.

Statement is FALSE. □

Part III Solutions: Counterexamples

Counterexample Gallery: Strict Separation of Classes



(a) **Strictly convex but not strongly convex:** $f(x) = x^4$.

$f''(x) = 12x^2 \geq 0$ (convex), and f is strictly convex because $f''(x) = 0$ only at the single point $x = 0$ (so f is not affine on any interval). However, $f''(0) = 0$, so there is no $\mu > 0$ with $f''(x) \geq 2\mu$ for all x . Hence f is not strongly convex.

(b) **Convex but not strictly convex:** $f(x) = |x|$ (or any affine function $f(x) = ax + b$).

Affine functions satisfy convexity with equality: $f(\lambda x_1 + (1 - \lambda)x_2) = \lambda f(x_1) + (1 - \lambda)f(x_2)$. Since equality holds for $x_1 \neq x_2$, f is not strictly convex.

(c) **Pseudoconvex but not convex:** $f(x) = -e^{-x}$.

$f''(x) = -e^{-x} < 0$ (strictly concave, so *not convex*). But $f'(x) = e^{-x} > 0$ for all x , so f is strictly increasing. Check pseudoconvexity: $f'(x_1)(x_2 - x_1) \geq 0$ implies $x_2 \geq x_1$ (since $f'(x_1) > 0$), which implies $f(x_2) \geq f(x_1)$ (strictly increasing). So f is pseudoconvex.

(d) **Strictly quasiconvex but not pseudoconvex:** $f(x) = x^3$.

$f'(0) = 0$ but $f(0) = 0$ is not a global minimum ($f(-1) = -1 < 0$). Since pseudoconvex functions have the property that every stationary point is a global minimizer, f is not pseudoconvex. However, f is strictly increasing as a function from $\mathbb{R} \rightarrow \mathbb{R}$ (even though $f'(0) = 0$, it doesn't change sign), so if $f(x_1) \neq f(x_2)$, the interior of $[x_1, x_2]$ stays strictly below the max. Hence f is strictly quasiconvex.

(e) **Quasiconvex but not strictly quasiconvex:** $f(x) = \min(x, 1)$.

The level sets $\{x : f(x) \leq \alpha\}$ are half-lines (convex), so f is quasiconvex. But take $x_1 = 0$ and $x_2 = 2$: $f(0) = 0$, $f(2) = 1$, and $f(\frac{1}{2} \cdot 0 + \frac{1}{2} \cdot 2) = f(1) = 1 = \max\{0, 1\}$, which is *not* strictly less than the max. So f fails the strict quasiconvexity condition.

Part IV Solutions: True/False

- (a) **True.** A strongly convex function f satisfies $f(\mathbf{x}) \geq f(\mathbf{x}^*) + \mu\|\mathbf{x} - \mathbf{x}^*\|^2$, which forces $f(\mathbf{x}) \rightarrow \infty$ as $\|\mathbf{x}\| \rightarrow \infty$. The minimizer exists (by coercivity) and is unique (by strict convexity).
- (b) **False.** Counterexample: $f(x) = x^3$ is quasiconvex, $f'(0) = 0$, but $x = 0$ is a saddle point (not a minimizer). For quasiconvex functions, stationary points can be saddle points or local maxima.
- (c) **False.** Counterexample: $f(x) = x^2$ is pseudoconvex, but $-f(x) = -x^2$ is pseudoconcave, not pseudoconvex. (A function that is both pseudoconvex and pseudoconcave is called *pseudolinear*.)
- (d) **False.** Counterexample: $f(x) = x$ and $g(x) = -x$ are both quasiconvex (they are linear). But $h(x) = f(x) + g(x) = 0$ is quasiconvex. Actually, this is a degenerate example. A better one: $f(x) = \max(x, 0)$ and $g(x) = \max(-x, 0)$ are both quasiconvex, but $f(x) + g(x) = |x|$ is also quasiconvex. To find a true counterexample, use two variables: let $f(x_1, x_2) = x_1$ and $g(x_1, x_2) = x_2$ (both quasiconvex). Then $h = f + g = x_1 + x_2$ is still quasiconvex. In general, the sum of quasiconvex functions is NOT necessarily quasiconvex, but constructing a clean 1D counterexample requires non-monotone functions. Consider $f(x) = \mathbf{1}_{x \geq 0}$ (indicator, quasiconvex) and $g(x) = \mathbf{1}_{x \leq 0}$, then $f + g$ has a local max at 0, breaking quasiconvexity. **The statement is FALSE** in general, even though many simple examples happen to work.
- (e) **True.** Linear functions are convex (and concave), hence pseudoconvex (by Part II(a)), and quasiconvex (convex $\Rightarrow$ QC). In fact, linear functions are *pseudolinear* (both pseudoconvex and pseudoconcave).
- (f) **True.** If f is strictly quasiconvex and $\bar{\mathbf{x}}$ is a local minimizer, suppose there exists $\mathbf{x}^* \neq \bar{\mathbf{x}}$ with $f(\mathbf{x}^*) \leq f(\bar{\mathbf{x}})$. If $f(\mathbf{x}^*) < f(\bar{\mathbf{x}})$: by strict quasiconvexity, $f(\lambda\bar{\mathbf{x}} + (1 - \lambda)\mathbf{x}^*) < f(\bar{\mathbf{x}})$ for $\lambda \in (0, 1)$, contradicting local minimality. If $f(\mathbf{x}^*) = f(\bar{\mathbf{x}})$: choose any $\mathbf{y}$ with $f(\mathbf{y}) < f(\bar{\mathbf{x}})$ (if no such $\mathbf{y}$ exists, $\bar{\mathbf{x}}$ is already global min). Then f must have value $< f(\bar{\mathbf{x}})$ on the open segment $(\mathbf{x}^*, \mathbf{y})$ by strict QC, contradicting local minimality near $\bar{\mathbf{x}}$. Hence $\bar{\mathbf{x}}$ is the unique global minimizer.